

Refining Research Questions & Deliverable Ideas

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Check-In/Turn and talk:



- Where are you in the process?
- Do you need to slow down? Speed up?
- How comfortable do you feel with your project?
- For groups: Have you been communicating with your group?



Deliverable Introduction



Show the world
your work!

Required: Poster for our
program Symposium



Research Questions

 As a group/independently:
Reflect on your research question

- What could be improved?
 - **How** can you do this?
- What is missing?
 - **How** can you fill that gap?



Turn and Talk!



Turn to the group next to you and share research questions. **GIVE** and **RECEIVE** 3 pieces of feedback (constructive criticism!)





Refining Research Questions

- 
- Take out your current research question. How specific is it? Does it include:
- Population impacted?
 - Specific treatment groups (or prototype)
 - Can you move beyond a question with a yes/no answer?
 - Could molecule A be replaced with molecule B in when creating compound C? → You figure out yes/no by experiment, but what if you add:
 - **If so, what implications exist for the cost and longevity of the final product? Think about WHAT your study brings to the table**

Tips + Self-Assessment

Here are two common issues with research questions, check to see if you can improve on these things:

- **You can answer the question too easily**
 - You can look it up and find the answer
 - One article alone answers this already
- **No relevant facts exist**
 - Question is based on opinions you feel sound right, not facts
 - Your question has already been disproven (find a new route!)
 - There are no direct/close sources to what you want to study (difficult to find supporting evidence for methods)

Activity: Research Paper Comprehension



- **Task:** We will read out loud the abstract and introduction of a paper. Your role is to find the answers to:
 - What they studied
 - What they cited/what other scientists found (hint: there is a citation next to each line taken from a different article)
 - **CHALLENGE:** Write a corresponding research question for the study (“challenge” but we want everyone to TRY this)



Activity: Our tips



- Start with identifying independent and dependent variables you see from study
 - HINT: there may be multiple! It is common for scientists to run various tests on same individuals
- TAKE NOTES of language you don't know to search up later
- Peek at the Materials and Methods section to learn more about WHAT they are doing (that will help you also identify variables and be more specific)



Activity: Our tips



Potential Research Question: Is there a correlation between the weekly gaming time (**one variable specified**) of Australian University students (**population specified**) and their assessed physical activity levels as recorded through an International Physical Activity Questionnaire-Short Form (**second variable and method specified**)?



Fun Activity!





Fun Activity – Oreo Separation

TASK: Each group gets will get TWO oreos, and you will create a research question to investigate the best way to seperate oreos cleanly (no cream on one side of oreo when pealed)

ex: Does peeling off the oreo fast or slow seperate it more cleanly?

STEPS:

1. With your group, come up with a research question (don't touch oreos yet!)
2. Conduct your experiment
3. Evaluate your results (which method worked better for the end goal?)



Fun Activity!



Was your research question **specific** (did you specify speed of removal, technique used, tools used)?

Did you have an instinctual answer to your research question before experimenting? Did this alter your interpretation of the results (this is something to avoid!)

What would you do differently next time?



Thank you!
